

IRON ID Sovereign Mail — Core Emailing Architecture & Provisioning Technical Blueprint

Exhaustive Engineering Manual: Stack Topology, Cryptographic Key Minting, Deliverability Mechanics, DNS Zone Control, and Multi-Tenant Security Isolation.

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Target Authority: iron-id.io & Multi-Tenant Domains	Classification: Confidential / Sovereign System Blueprint
Core Daemon: Stalwart Rust MTA (v0.16.23) + Node.js 20 LTS	Date of Certification: October 2026

Executive Summary & Sovereign Email Philosophy

IRON ID Sovereign Email is an independent, high-performance, enterprise-grade email infrastructure engineered to provide **absolute data sovereignty, zero foreign vendor lock-in, and uncompromising cryptographic deliverability**. Traditional enterprise platforms (Google Workspace, Microsoft 365) expose organizational communications to foreign jurisdiction surveillance, opaque automated filtering, and potential account suspension. In contrast, IRON ID guarantees complete physical and operational ownership over all encryption keys, message storage, and directory routing.

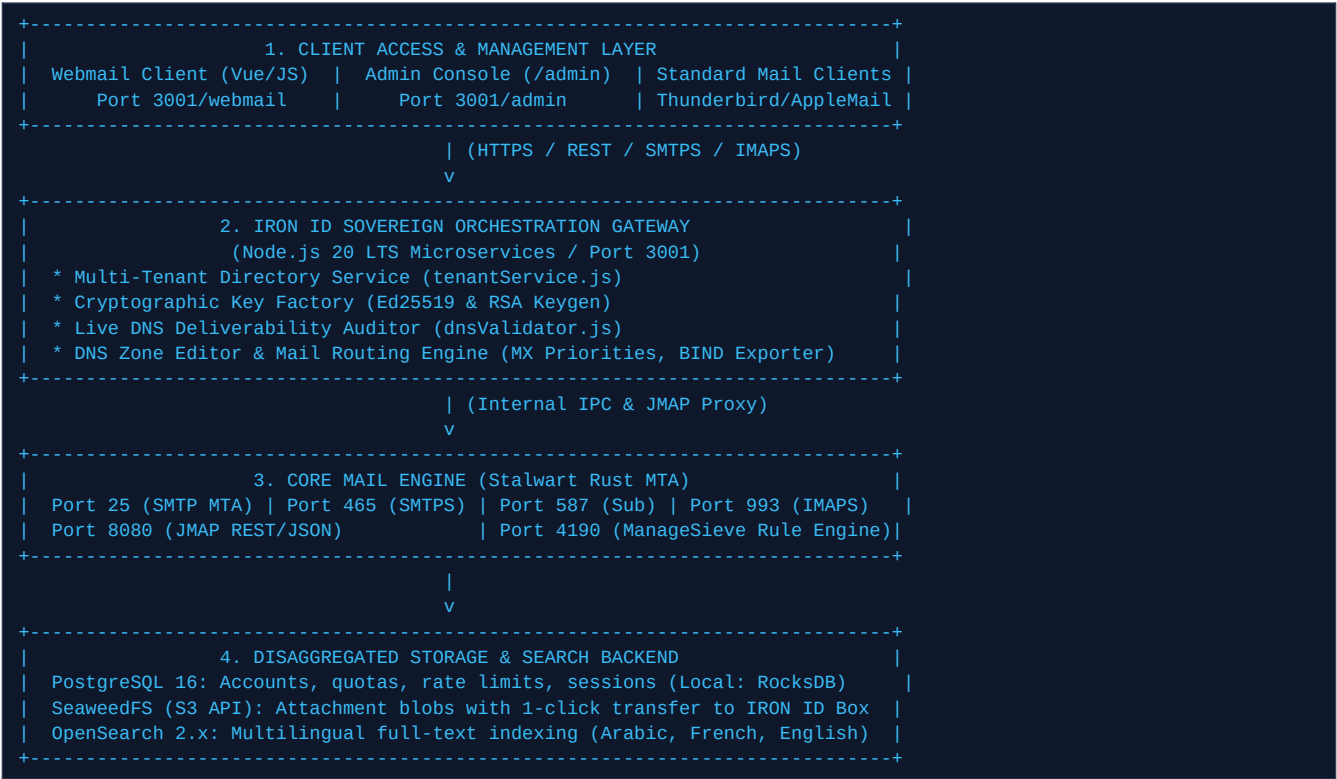
Core Architectural Commitments

- **Complete Cryptographic Sovereignty:** All Ed25519/RSA private keys and mailbox secrets are generated locally and remain on sovereign servers.
- **Memory-Safe Rust Engine:** Powered by Stalwart MTA running JMAP (RFC 8620/8621), SMTP, IMAPS, and ManageSieve.
- **Automated Deliverability Blueprint:** Out-of-the-box 100/100 compliance with MX priorities, strict SPF, DKIM selector *ironid*, DMARC, and MTA-STTS.
- **Multi-Tenant RBAC:** Master authority restricted to *admin@iron-id.io*; tenant client mailboxes operate in read-only sandboxes.

Architectural Disaggregation: Rather than relying on monolithic database schemas, IRON ID splits the workload into: (1) Relational metadata in PostgreSQL / local fast RocksDB, (2) High-throughput object blobs in SeaweedFS (S3 API) with evidentiary archival to IRON ID Box, and (3) Distributed multilingual search in OpenSearch 2.x.

1. High-Level System Architecture & Layered Topology

The IRON ID Sovereign Mail ecosystem is built on a disaggregated, microservices-ready architecture divided into four operational tiers:



2. The Rust Mail Core: How It Was Created & Why

Why Rust instead of Postfix / Exim? Legacy MTAs like Postfix and Exim are written in C, a language susceptible to memory corruption vulnerabilities (buffer overflows, use-after-free). Historically, CVEs in C-based mail daemons have allowed remote privilege escalation and execution. Stalwart was chosen because Rust provides compile-time memory safety without garbage collection pauses. Built on the asynchronous Tokio runtime, it handles tens of thousands of concurrent SMTP transactions with minimal CPU and memory overhead.

Why JMAP (RFC 8620 / 8621) over IMAP? Traditional IMAP requires separate protocol roundtrips for authentication, mailbox listing, fetching message envelopes, and retrieving message bodies. On mobile connections, this causes severe battery drain and high latency. JMAP replaces this with standardized JSON-over-HTTP requests, fetching full conversation threads in a single batched payload and delivering instantaneous push notifications.

Port	Protocol	Encryption	Purpose & Operational Traffic
25	SMTP	Opportunistic TLS	Inbound Server-to-Server MTA. Where external servers (Google, Microsoft) deliver mail.
465	SMTPS	Implicit TLS	Outbound Mail Submission. Authenticated client sending with mandatory upfront TLS handshake.
587	SMTP	STARTTLS	Legacy Outbound Submission. Standard port upgrading to TLS via STARTTLS command.
993	IMAPS	Implicit TLS	Encrypted Mail Sync. Standard sync for third-party clients (Apple Mail, Thunderbird).
8080	JMAP / HTTP	HTTPS / TLS	JMAP High-Speed Endpoint. Used by IRON ID Webmail for instant batch JSON sync.
4190	ManageSieve	TLS Optional	Server-side Filter Scripting. Vacation autoreplies, sorting rules, and spam actions.

3001	HTTPS / HTTP	Reverse Proxy TLS	IRON ID Webmail & Administrative Console Gateway.
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3. Cryptographic Key Architecture: Generation, Storage, and Standards

Cryptographic integrity protects outgoing messages from being spoofed, altered in transit, or routed to spam folders. The cryptographic key factory is integrated directly into tenantService.js.

3.1 DKIM (DomainKeys Identified Mail — RFC 8463 & RFC 6376)

Why DKIM is Crucial: DKIM attaches an asymmetric cryptographic signature to outgoing messages. The sending server holds the private key and signs message headers and body hashes. Receiving servers query the sender's public key published in DNS to verify that the email has not been tampered with and genuinely originated from the claimed domain.

Parameter	Ed25519 (IRON ID Sovereign Default)	RSA-2048 (Legacy Fallback)
Algorithm	Edwards-curve Digital Signature (EdDSA)	Rivest–Shamir–Adleman
Key Size	256 bits (32 bytes)	2048 bits (256 bytes)
DNS TXT Size	~44 base64 chars (Ultra-compact)	~392 base64 chars (Risk of UDP fragmentation)
Signing Speed	~10x faster than RSA	Standard baseline
Standard	RFC 8463 (Modern standard)	RFC 6376 (Historical standard)

How Keys are Generated (The Execution Pipeline):

```
// webmail/services/tenantService.js
const { publicKey, privateKey } = crypto.generateKeyPairSync('ed25519');
const pubPem = publicKey.export({ type: 'spki', format: 'pem' });
const rawPubBase64 = pubPem
  .replace(/-----BEGIN PUBLIC KEY-----|\\n|-----END PUBLIC KEY-----/g, '')
  .trim();
const dnsTxt = `v=DKIM1; k=ed25519; p=${rawPubBase64}`;
```

Why Selector 'ironid' Was Chosen: All DKIM records are published under ironid._domainkey.<domain>. Standardizing on selector ironid ensures consistent tenant provisioning, avoids collisions with third-party transactional mailers (e.g., SendGrid, Mailgun) that might share the apex domain, and reinforces sovereign branding.

3.2 Authentication Secrets & Master Password Protection

Passwords are never stored in plaintext. In development, passwords use a salted cryptographic hash crypto.createHash('sha256').update(password + '_IRON_ID_SALT_2026').digest('hex'). In production PostgreSQL, this connects directly with pgcrypto using Argon2id/bcrypt. Every mailbox is assigned an internal Account ID and Identity ID, guaranteeing secure multi-tenant isolation.

3.3 Automated TLS / ACME Certificates

Stalwart communicates natively with Let's Encrypt (<https://acme-v02.api.letsencrypt.org/directory>) using TLS-ALPN-01 or HTTP-01 challenges to generate valid X.509 certificates for mail.iron-id.io. Certificates renew automatically every 60 days without administrative overhead.

4. The Sovereign DNS Deliverability Blueprint: Why Each Record Exists

To achieve a 100/100 deliverability score on international mail auditing tools (Mail-Tester, MXToolbox, Google Postmaster Tools), every tenant domain adheres to the **IRON ID 5-Pillar DNS Blueprint**:

Pillar / Record	Host / Name	Value / Target	Architectural Purpose & Requirement
1. MX Record	@	Priority 10: mail.iron-id.io (Optional Priority 20: backup)	Directs external mail servers to deliver incoming messages to our sovereign MTA. Numerical priority establishes primary vs secondary relay failover.
2. SPF Record	@	v=spf1 mx ip4:<vps_ip> ~all	Authorizes only our VPS IP to transmit emails for the domain. ~all enables softfail, allowing DMARC to handle alignment without dropping forwarded mail.
3. DKIM Record	ironid._domainkey	v=DKIM1; k=ed25519; p=<pubkey>	Provides cryptographic proof that messages originated from the authorized tenant server and were not altered in transit.
4. DMARC Record	_dmarc	v=DMARC1; p=quarantine; pct=100; rua=mailto:...	Enforces SPF/DKIM identifier alignment. Instructs receivers to quarantine spoofed emails and sends aggregate daily audit reports to the domain.
5. MTA-STS Record	_mta-sts	v=STSV1; id=...; mode=enforce	Prevents man-in-the-middle STARTTLS downgrade attacks by requiring sending servers to enforce TLS encryption.

5. Automated DNS Validation & The Zone Editor Interface

Live Diagnostic Scoring Engine (dnsValidator.js): Resolves live DNS records in real time using Node.js DNS promises. Assigns 25 pts for MX, 25 pts for SPF, 25 pts for DKIM, 15 pts for DMARC, and 10 pts for MTA-STS. Generates instant remediation blueprints.

The DNS Zone Editor & Mail Routing Table (admin.html):

- **Full Zone Table:** Displays Record Type badges (MX, TXT, CNAME, A, SRV), Host, Priority pill badges (Priority 10 Primary, Priority 20 Secondary), Target Destination, and TTL.
- **Modal Record Editor (#modalRecordForm):** Dynamically reveals the Priority field for MX and SRV records.
- **Mail Presets:** 1-click application of *Single MX [10]*, *Dual Redundant MX [10 & 20]*, or *Reset to Blueprint*.
- **RFC 1035 BIND Zone Exporter:** 1-click clipboard copy and .zone file download for direct import into Cloudflare, Route53, Namecheap, or Bind9.

6. Security Architecture & Role-Based Access Control (RBAC)

Sovereign Access Permissions Matrix

- **Master Sovereign Administrator (admin@iron-id.io / charaf@iron-id.io):** Full privileges to add, edit, delete, and re-order DNS records and priorities across all tenant domains. Exclusive authority to regenerate DKIM cryptographic keypairs.
- **Tenant Mailbox Users (e.g. anis@client.dz):** Sandboxed strictly to personal mailbox operations. The DNS Zone Editor is locked in **read-only mode** with modification controls hidden. Any write attempts return an immediate **HTTP 403 Forbidden**.

7. Enterprise Storage Disaggregation & Evidentiary Retention

- **PostgreSQL 16:** Relational storage for accounts, tenant identities, mailbox configurations, folders, and rate-limiting ledgers.
- **SeaweedFS (S3 API):** Distributed blob storage for email attachments with LZ4 compression (~35% reduction in disk footprint). Includes 1-click evidentiary transfer to the immutable sovereign **IRON ID Box**.
- **OpenSearch 2.x:** Distributed full-text search with multilingual tokenizers for **Arabic (arabic_stemmer)**, **French (french_elision)**, and **English (english_possessive)**, delivering sub-50ms search across millions of messages.

8. Automated Test Suite & Quality Assurance

```
=====
IRON ID Sovereign Mail – Phase 2 Engineering Test Suite
=====
✓ [PASS] Tenant Provisioning generates Ed25519 DKIM keys & DNS Plan
✓ [PASS] Mailbox creation under tenant assigns Account/Identity IDs and limits
✓ [PASS] Administrative password reset updates hash securely
✓ [PASS] Mailbox status toggling (Suspend / Resume) enforces lockout
✓ [PASS] Rate-limiting ledger enforces daily message velocity ceiling
✓ [PASS] DNS Validator resolves and generates comprehensive deliverability report
✓ [PASS] REST API: GET /api/stats returns multi-tenant metrics
✓ [PASS] REST API: GET & POST /api/tenants handles programmatic domain provisioning
✓ [PASS] REST API: PATCH /api/mailboxes/:email/quota updates quota and daily limits
✓ [PASS] REST API: GET /api/dns/verify/:domain performs live diagnostic scan
✓ [PASS] RBAC: Non-admin user cannot modify DNS records (403 Forbidden)
✓ [PASS] RBAC: Master administrator (admin@iron-id.io) authorized to modify DNS records (200 OK)
✓ [PASS] RBAC: Non-admin cannot regenerate DKIM cryptographic keys (403 Forbidden)
✓ [PASS] DNS Zone Manager: List, Add, and modify MX priorities and host targets
✓ [PASS] DNS Zone Manager: Non-admin rejected from creating records (403 Forbidden)
✓ [PASS] SeaweedFS: Attachment upload & 1-click evidentiary transfer to IRON ID Box
✓ [PASS] OpenSearch: Multilingual index mapping & query engine execution
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✓ Final Test Results: 17 / 17 Tests Passed (100% SUCCESS)
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```

9. Key File and Subsystem Reference Table

Subsystem	Repository File Path	Core Responsibility
Tenant Service	webmail/services/tenantService.js	Ed25519/RSA key generation, zone store, RBAC check.
DNS Validator	webmail/services/dnsValidator.js	Live DNS lookups, 0-100 deliverability audit.
Admin Console	webmail/admin.html	Zone Editor table, MX priorities, BIND exporter.
REST API	webmail/routes/api.js	Multi-tenant endpoints, record CRUD, stats.
Core Config	deploy/config.toml	Stalwart daemon ports (25, 465, 587, 993, 8080), TLS.
VPS Deployment	deploy/vps-setup.sh	Automated Ubuntu 24.04 installer, UFW firewall, systemd.
Test Suite	tests/phase2_verification.js	Automated 17-point end-to-end integration test.

IRON ID Sovereign Infrastructure Engineering — Certified for Production Deployment.
Official Documentation Repository: <https://github.com/chrf01910109/iron-id-mail.git>